

Supporting Parental Online Content Mediation in the Era of GenAI through Assistive Caregiving

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Abstract

As children engage with online multimedia content across different developmental stages, it is increasingly difficult for parents or caregivers to ensure children's online safety while balancing their autonomy and privacy. Through my doctoral research, I examine if (and how) generative AI (GenAI) can function as an assistive intelligence supporting collaborative decision-making between caregivers and children to help them mediate children's online multimedia experiences. My research seeks to determine: (1) how GenAI can support parents and caregivers in evaluating and contextualizing online multimedia content for young children (ages 0–7), and (2) how GenAI can enable collaborative content supervision between parents/caregivers and older children (ages 9–12) while respecting their privacy as children transition toward increased independence.

CCS Concepts

• Social and professional topics → Children.

Keywords

Online Child-Safety, GenAI, Parents, Children, Collaborative Content Moderation

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1 Introduction

Online multimedia content are an indispensable part of childhood for the children born in the age of information and technology [5]. More than half of children in the United States have their own mobile device by age eight [8], and a large majority of parents report concerns about the extent and nature of their children's exposure to online multimedia content [6]. As the volume and diversity of online multimedia content continue to grow [23], parents and caregivers are finding it increasingly challenging to regulate children's

exposure to online multimedia content. While such exposure provides opportunities for children to explore new topics, encounter diverse perspectives, and access entertainment with ease [15], it also introduces the significant risks of inadvertently encountering inappropriate or disturbing content [17]. Unlike traditional media that adhere to structured distribution models and well-established rating systems (e.g., Motion Picture Association ratings¹ or TV Parental Guidelines²), online platforms often lack consistent and standardized age-based moderation mechanisms, which results in parents and caregivers bearing the primary responsibility for screening content [1, 15].

Parents and caregivers have adopted several mediation strategies like Active Mediation,³ Restrictive Mediation,⁴ Co-viewing,⁵ and more recently, Joint Media Engagement (JME)⁶. These mediation practices vary across different developmental stages. While younger children (ages 0–7) are largely dependent on caregivers for content moderation, older children (ages 9–12) transitioning to adolescence increasingly seek independence [11]. With society entering the era of intelligent systems [13], AI-driven assistive technologies open up new opportunities in supporting families navigate the complex online environments in an explainable and collaborative manner.

Through my doctoral research, I seek to establish a grounded understanding of how generative AI (GenAI) can assist parental decision-making and contribute conceptual insights to inform the development of privacy-sensitive, developmentally appropriate approaches towards a GenAI-based assistive caregiving intermediary to support caregivers in mediating children's online multimedia experiences. My work investigates how GenAI can help parents evaluate and contextualize online content for younger children (ages 0–7), and how it can facilitate collaborative content supervision with older children (ages 9–12) as they become more independent. I am particularly interested in examining how GenAI can combine automated assistance with human judgment to provide trustworthy, transparent, and privacy-sensitive support for families to enable a safer approach to online content supervision without relying on restrictive or pervasive surveillance.



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¹MPA: <https://www.motionpictures.org/film-ratings/>

²TV Parental Guidelines: <https://www.tvguidelines.org/>

³Active Mediation: Adults engaging in discussions about media content with their children [19–21]

⁴Restrictive Mediation: Adults limiting children's access to online media [20, 21]

⁵Co-viewing: Adults and children sharing the viewing experience without discussions about the content [20]

⁶Joint Media Engagement (JME): Adults sharing the viewing experience (co-viewing) and actively engaging in meaningful discussion [18]

2 Proposed Dissertation Work

My initial work explores the capabilities of GenAI in moderating and contextualizing multimedia content for young children (0-7 years) using both quantitative and qualitative research methodologies. Based on the results of the exploration, I plan to design a privacy-aware GenAI-based assistive caregiving intermediary for older children (9-12 years). Figure 1 shows how the initial work connects with the ongoing work and proposed research plan.

2.1 Exploring How GenAI Can Assist Parents and Caregivers in Evaluating Content Appropriateness for Young Children

Children under the age of eight are in a developmental stage where self-regulation skills and media literacy are still maturing. It makes them largely dependent on parents or caregivers to regulate their online exposure [15]. Prior research most widely recommends Joint Media Engagement (JME) as the best strategy for regulating younger children's exposure [10, 24], but the scale and diversity of online video platforms make it increasingly difficult for caregivers to actively engage (co-viewing and discussing) with the content. This leads to dependency on existing automated moderation systems that often lack transparency and fail to reflect nuanced parental values. In my first work, I examined whether large language models⁷ (LLMs) can help bridge this gap. I sought to understand how caregivers of young children (ages 0–7) assess the appropriateness of online multimedia contents by answering the research questions:

- RQ1** How well can LLMs detect and describe the appropriateness of YouTube videos for children under the age of 7? How do LLMs reason about appropriateness?
- RQ2** How well do LLMs and parental reasoning align? How can the LLM aid parents in making decisions more effectively?

We conducted two consecutive online survey-based studies to answer the research questions. Our first study evaluated the LLM's child-appropriate video detection and explanation capabilities by recruiting 120 parents as experts, and qualitatively analyzed the LLM-generated explanations to understand the rationale provided to back those decisions. Participants in this study were asked to assess content independently and then review the LLM description about the appropriateness of the content for young children. We analyzed the parents' rationale for their moderation decisions from their independent responses and conducted the second study to evaluate the incorporation of the parental preferences in the LLM-generated descriptions. We recruited 250 parents to evaluate the effectiveness of two distinct sets of descriptions designed with the LLM as a co-designer, and analyzed the parents' responses quantitatively by statistically comparing the alignment of the decisions made by parents after viewing the full video with their decision solely based on the LLM-generated descriptions.

Findings from the studies indicate that while LLMs are not reliable content moderators by themselves, they can provide valuable contextual explanations that support parental judgment. Parents make statistically equivalent decisions to watching the entire video

when they make decisions based on the descriptions that incorporate parental values. They showcased a high level of satisfaction with the descriptions and reported that GenAI-generated descriptions helped them better understand the content, identify potential risks, and helped them articulate their own concerns. Our studies show that positioning GenAI as an assistive tool rather than an authoritative decision-maker supported more informed mediation practices, and this approach has the potential to help parents initiate JME conversations. It highlights the potential of GenAI as a supportive intermediary that enhances parental expertise without replacing human judgment. This work is accepted for publication in CHI '26 [14].

2.2 Parental Concerns & Expectations regarding GenAI Caregiving Assistants for Collaborative Content Moderation (Ongoing)

Children seek greater independence as they grow up [11]. Prior research shows that open communication and active mediation are associated with reduced online risks for this age group [4, 7, 9] while restrictive or interventionist approaches limit opportunities for them to develop effective coping strategies [22] and prompt them to develop mitigation strategies to preserve their autonomy and resist supervision systems [3]. Our first work demonstrates the potential of GenAI as an assistive technology, but it also raises important questions about the design of such systems to align with parents' expectations, values, and concerns when moderating content for older children. When we evaluated GenAI for assisting in content moderation for young children, parents and caregivers were the only individuals directly interacting with the system. However, supporting collaborative mediation for older children requires GenAI-based assistive caregiving tools to account for children's engagement with AI, and given parents' emerging concerns about children's exposure to AI [25], I aim to explore parental concerns about the AI exposure and their expectations from a safe and privacy-aware GenAI-based caregiving assistant in the second phase of my dissertation.

In this phase, we are conducting in-depth semi-structured interviews with parents. Our study is focusing on: (1) parents' attitudes and acceptability toward an "AI Caregiver Assistant" (e.g., an AI mediator or guide) in supporting their content moderation practices; (2) parents' perceptions of the level of autonomy they are comfortable granting children and their concerns about children's safety and exposure to AI in the context of AI-assisted content moderation; and (3) the level of specificity in AI-generated descriptions of online multimedia content that parents desire to support effective content moderation while respecting children's privacy. This work seeks to understand how parents reason about trade-offs between effectiveness, safety, and their children's autonomy and privacy when AI systems are involved in content moderation, and how they envision GenAI integrating into their existing mediation practices. This is a work in progress.

We expect the findings from this study to clarify how parents articulate their concerns and boundaries around AI involvement in children's online content moderation with respect to children's safety, privacy, and autonomy. This phase will build directly on the

⁷Large Language Models (LLMs) are a subset of GenAI that focuses on text-generation [16]

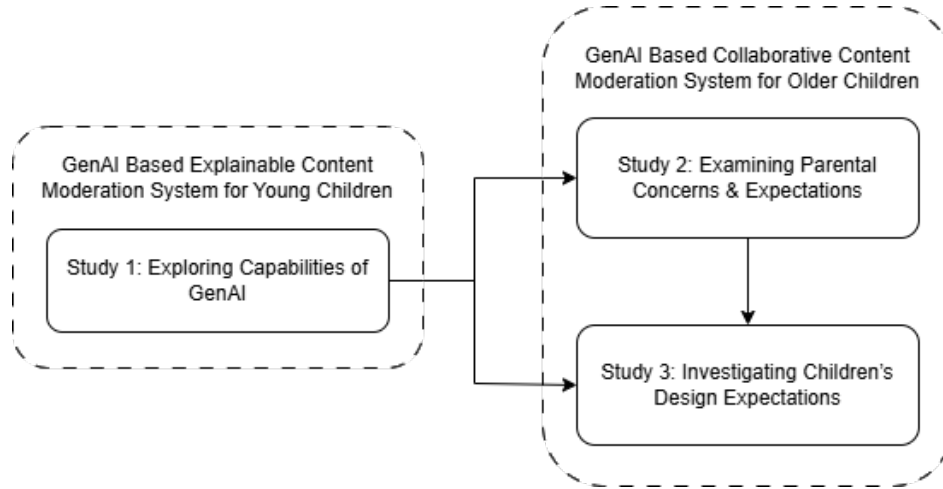


Figure 1: Proposed Dissertation Work Flow

first study by expanding the scope to older children and exploring how they want GenAI-based caregiving assistants to behave and integrate into their existing mediation practices.

3 Research Plan

While my ongoing work focuses on the opinions and concerns of the parents about collaborative content supervision systems involving AI, it is extremely important to include the children in the design process to ensure that they feel comfortable with their privacy and autonomy. In the next part of my dissertation, I plan to examine children's perspectives on AI-assisted collaborative content supervision. The study will investigate:

- (1) How do children perceive AI-based monitoring and intervention in comparison to direct parental involvement?
 - What are their concerns about the involvement of AI versus human caregivers in overseeing their online activities?
 - Would they be more comfortable with intervention-based approaches versus monitoring (also active vs passive monitoring)?
- (2) What are the privacy-preserving design expectations from children for the AI-based collaborative content moderation system?
 - Which details are the children comfortable sharing?
 - Which details do they prefer to keep private?
 - How do these preferences vary across different aspects of online content (e.g., topic, channel, or influencer identity, language use)?

To answer the questions, we plan to either:

- Recruit and interview children aged 9 to 12 (one-to-one) in a semi-structured manner. We can hold the interviews in a home setting or elsewhere. This approach allows us to gain a better understanding of each child's concerns and opinions.
- Host a workshop-style interview with a group of children aged between 9 and 12. We can provide them with activities and observe how they interact with the AI system. It can be designed to be a co-design workshop where we flesh out the

design expectations implicitly from the children's actions. The children might be less likely to share details in a group setting, but this will allow them to be more comfortable in their own settings.

We are finalizing the details of this study. I hope to build on insights from the second study to combine it with the older children's (aged 9 to 12) expectations for privacy-preserving design in GenAI-based assistive caregiving systems to identify the design principles that respect children's emerging autonomy while maintaining appropriate safeguards.

4 Expected Contributions

The results from my first work:

- Reinforces longstanding HCI arguments for human-in-the-loop approaches [2, 12] by demonstrating how GenAI can function as an assistive tool that enhances parental expertise rather than replacing parental judgment.
- It shows that GenAI-generated explanations can simplify informed parental decision-making and support Joint Media Engagement (JME) behaviors by enabling parents and children to discuss online content meaningfully through contextual descriptions, even when co-viewing the content in its entirety is impractical.

The second study is expected to provide a deeper understanding of parents' perceptions, concerns, and expectations regarding GenAI-based assistive mediation for content moderation by building on the findings from the first work. It will contribute to:

- Clarifying how parents reason about appropriate levels of AI involvement and description when using GenAI to support content supervision.
- Designing a more explainable content mediation system that respects the parental concerns about exposure to AI systems while effectively aiding them in the collaborative supervision process.

Together with the planned children's study that explores the children's design expectations, this dissertation will contribute to

a more holistic understanding of how privacy-aware GenAI caregiving assistants can support developmentally appropriate content supervision. These contributions will inform HCI research on assistive AI, child-centered technologies, and the design of AI systems that engage families in sensitive, high-stakes decision-making contexts.

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